LOVELY PROFESSIONAL UNIVERSITY

Sample Question Set - Unit 5

Course Code: CSE276
Course Title: ARTIFICIAL INTELLIGENCE FOUNDATIONS
Course Outcome: CO5
Paper: Sample Multiple Choice Questions - ANSWER KEY
Maximum Marks: 40 x 1 = 40

Unit: 5
Unit Title: Generative AI, Responsible and Explainable AI
Unit Mapped RBT: L2 (Understand)
Total Questions: 40
Suggested Duration: 60 Minutes

Answer Grid (Key)

Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
Q1	A	Q11	D	Q21	A	Q31	C
Q2	B	Q12	B	Q22	C	Q32	A
Q3	C	Q13	D	Q23	B	Q33	A
Q4	D	Q14	C	Q24	D	Q34	D
Q5	B	Q15	B	Q25	C	Q35	B
Q6	D	Q16	A	Q26	D	Q36	C
Q7	A	Q17	B	Q27	A	Q37	C
Q8	C	Q18	A	Q28	B	Q38	B
Q9	C	Q19	D	Q29	D	Q39	D
Q10	A	Q20	C	Q30	B	Q40	A

Multiple Choice Questions (40 x 1 = 40 Marks)

Q1. Name the class of model that produces new content rather than assigning a label to an input.

A) A generative model <-- correct
B) A discriminative model
C) A production system
D) A search algorithm

Answer: A. A generative model composes text, images or code. A discriminative model only separates the inputs it is shown into categories.

Q2. Identify the limit on how much text a large language model can take into account at one time.

A) The temperature setting
B) The context window <-- correct
C) The token vocabulary
D) The parameter count

Answer: B. Everything the model can use, including the system prompt and any retrieved passages, has to fit inside the context window.

Q3. State what the role instruction of a prompt supplies to the model.

A) The maximum length of the reply
B) The documents it should search
C) The character or expertise it should adopt <-- correct
D) The temperature it should use

Answer: C. Setting a role, such as an examiner or a tutor, shifts the register and the depth of the answer before any other instruction is read.

Q4. Name the prompting technique that includes a few worked examples inside the prompt.

A) Zero-shot prompting
B) Chain-of-thought prompting
C) Retrieval grounding
D) Few-shot prompting <-- correct

Answer: D. The examples demonstrate the pattern of answer wanted, which is often more effective than describing that pattern in words.

Q5. Identify the first step in preparing a document collection for Retrieval-Augmented Generation.

A) Fine-tuning the model on the documents
B) Splitting the documents into passages <-- correct
C) Deleting every duplicate document
D) Translating the documents into English

Answer: B. Documents are chunked into retrievable passages, which are then embedded and indexed so that the relevant few can be fetched at question time.

Q6. State what a vector store holds in a retrieval pipeline.

A) The full text of every user question
B) The parameters of the language model

C) The rules of the organisation's policy
D) Numeric embeddings of the text passages <-- correct
Answer: D. Passages are stored as vectors so that a question can be matched by meaning rather than by exact wording.

Q7. Name the Responsible Artificial Intelligence principle that requires personal data to be protected.

A) Privacy <-- correct
B) Transparency
C) Accountability
D) Fairness
Answer: A. Privacy covers what is collected, how long it is kept, who can see it and whether it leaves the organisation at all.

Q8. Identify the Responsible Artificial Intelligence principle requiring that a named person be answerable for a decision.

A) Fairness
B) Privacy
C) Accountability <-- correct
D) Transparency
Answer: C. Accountability prevents the system itself being offered as the reason for an outcome; a person must remain answerable for it.

Q9. State the term for an attack in which text hidden inside an input changes the instructions a model follows.

A) Data poisoning
B) Model inversion
C) Prompt injection <-- correct
D) Denial of service
Answer: C. Because instructions and data arrive as the same stream of text, content the model reads can carry directions of its own.

Q10. Name the kind of explanation that accounts for one particular prediction.

A) A local explanation <-- correct
B) A global explanation
C) A statistical summary
D) A model card
Answer: A. A local explanation says why this applicant, with these values, received this outcome. A global one describes the model's behaviour overall.

Q11. Identify the term for a synthetic image or video made to depict a real person convincingly.

A) A hallucination
B) An embedding
C) A guardrail
D) A deepfake <-- correct
Answer: D. Deepfakes are among the societal risks of generative models, bearing on misinformation, fraud and consent.

Q12. State the term for data that identifies a particular individual.

A) Aggregated statistical data
B) Personally identifiable information <-- correct
C) Synthetic training data
D) Publicly licensed data
Answer: B. Names, identifiers, contact details and any combination of fields that singles out a person all fall under this heading.

Q13. Name the practice of collecting only the data actually required for a stated purpose.

A) Data augmentation
B) Data replication
C) Data labelling
D) Data minimisation <-- correct
Answer: D. Minimisation limits both the harm a breach can cause and the scope for the data to be used for something it was never gathered for.

Q14. Identify the setting that makes the output of a language model more predictable and less varied.

A) A high temperature
B) A larger context window
C) A low temperature <-- correct
D) A longer system prompt
Answer: C. Low temperature concentrates the choice on the most probable continuations, which suits factual work; high temperature suits creative drafting.

Q15. State what Artificial Intelligence governance provides within an organisation.

A) The hardware on which the models are trained
B) Policies, approval and oversight for the use of AI <-- correct
C) The training data used to build the models
D) The programming language used for the code
Answer: B. Governance decides what may be built, who signs it off, what must be documented and how a system in service is reviewed.

Q16. Name the generative application that reduces a long document to its principal points.

A) Summarisation <-- correct
B) Classification
C) Object detection
D) Clustering
Answer: A. Summarisation, translation, drafting and code generation are the standard generative applications named on the syllabus.

Q17. Explain the difference between a generative model and a discriminative one.

A) One separates inputs, the other stores them
B) One composes new content, the other separates inputs <-- correct
C) Both of the two compose entirely new content
D) Both of the two only separate inputs into classes

Answer: B. A spam filter decides which of two classes a message belongs to. A generative model writes a message that did not previously exist.

Q18. Distinguish a model that is interpretable by design from one explained after the event.

A) The first is readable in itself, the second is approximated <-- correct
B) The second is readable in itself, the first approximated
C) Both of the two are readable in themselves
D) Both of the two are approximated after training

Answer: A. A short decision tree can simply be read. A large network can only be explained by a separate method that estimates what influenced its output.

Q19. Describe what splitting documents into passages contributes to a retrieval pipeline.

A) It removes every factual error from the documents
B) It reduces the number of parameters in the model
C) It translates the documents into a common language
D) It lets only the relevant part be retrieved and supplied <-- correct

Answer: D. The context window is limited, so a whole manual cannot be supplied. Retrieving the few passages that bear on the question is what makes grounding possible.

Q20. Interpret a system that answers only from passages it has fetched and names the passage it used.

A) It is a model that has been fine-tuned on the data
B) It is a search engine with no model at all
C) It is a grounded, retrieval-augmented assistant <-- correct
D) It is a rule-based expert system with citations

Answer: C. Grounding restricts the answer to supplied source material and makes the source checkable, which is the principal defence against hallucination.

Q21. Predict what has to happen to a retrieval assistant when one of its source documents is revised.

A) The revised passage must be re-embedded and re-indexed <-- correct
B) The whole language model must be retrained
C) Nothing at all, since the model adapts by itself
D) The system prompt has to be rewritten fully

Answer: A. The model itself is unchanged; the store is what holds the content. This is exactly the advantage retrieval has over fine-tuning for material that changes.

Q22. Recognize the risk described: a web page an assistant is asked to read contains hidden text instructing it to disregard its own rules.

A) Hallucination of a source
B) Loss of the context window
C) Prompt injection <-- correct
D) Overfitting to the page

Answer: C. The model has no reliable way to separate instructions from content, so untrusted text it processes must be treated as potentially adversarial.

Q23. Explain why asking a model to set out its intermediate steps can improve a multi-step answer.

A) The model consults an external calculator
B) The steps become part of the context it reasons from <-- correct
C) The model is switched to a larger version
D) The model retrieves the answer from a database

Answer: B. Each step it writes conditions what it writes next, which keeps a long chain of reasoning on track instead of jumping to a conclusion.

Q24. Summarize the sources from which bias enters a deployed model.

A) The programming language and the hardware
B) The size of the context window in use
C) The number of users the system serves
D) The training data, the sampling and the labelling <-- correct

Answer: D. Historical decisions, unrepresentative samples and the judgements of the people who applied the labels all leave their mark on what the model learns.

Q25. Distinguish fairness from accuracy as objectives for a deployed model.

A) Fairness is overall correctness, accuracy concerns groups
B) Both of the two measure overall correctness
C) Accuracy is overall correctness, fairness concerns groups <-- correct
D) Both of the two measure performance by group

Answer: C. The two can pull in opposite directions, and a system can be highly accurate in aggregate while being systematically worse for one group.

Q26. Explain how a model can be accurate overall and still unfair to a particular group.

A) Accuracy is always computed on one group only
B) Fairness requires the accuracy to be exactly zero
C) The overall figure cannot be computed by group

D) A small group's errors barely affect the overall figure <-- correct

Answer: D. A group that is 3 per cent of the population can be served badly while the headline accuracy stays high, which is why results must be broken down by group.

Q27. Interpret an explanation that lists which features mattered most across the whole population of cases.

A) It is a global explanation <-- correct

B) It is a local explanation

C) It is a fairness audit

D) It is a privacy assessment

Answer: A. A global explanation describes the model's overall behaviour. A person affected by one decision needs the local explanation instead.

Q28. Predict the effect on a language model's output of raising its temperature setting.

A) The output becomes shorter and more factual

B) The output becomes more varied and less predictable <-- correct

C) The output becomes identical on every run

D) The output stops depending on the prompt

Answer: B. Higher temperature flattens the choice among probable continuations. It suits brainstorming and is a poor choice for anything that must be reliable.

Q29. Describe what a guardrail on a deployed assistant does.

A) It increases the speed at which answers appear

B) It expands the amount of text the model can read

C) It retrains the model whenever an error occurs

D) It blocks or redirects inputs and outputs outside policy <-- correct

Answer: D. Guardrails sit around the model, screening what goes in and what comes out, because the model itself offers no guarantee about either.

Q30. Explain why removing names from a dataset does not always remove the privacy risk.

A) The names are still stored inside the model file

B) Remaining fields in combination can still identify a person <-- correct

C) Anonymised data cannot be used to train a model

D) The risk transfers entirely to the model's output

Answer: B. A postcode, a date of birth and a job title together can pick out one individual. Re-identification from combinations is the standard difficulty.

Q31. Distinguish the training data of a model from the prompt it is given at the time of use.

A) A prompt shapes the model, training data one reply

B) Both of the two permanently change the model

C) Training data shapes the model, a prompt shapes one reply <-- correct

D) Both of the two affect only a single reply

Answer: C. A prompt changes nothing about the model's parameters. This is why prompt engineering is cheap and reversible while training is neither.

Q32. Explain why a generative model should not be the only source of a factual claim.

A) It is trained to produce fluent text, not verified fact <-- correct

B) It cannot produce any factual statement at all

C) It only repeats sentences from its training data

D) It refuses to answer any factual question

Answer: A. Fluency and correctness are separate properties. A confident, well-formed sentence carries no guarantee that the fact within it exists.

Q33. Recognize the mitigation described: the assistant declines to answer and passes the question to a person when retrieval returns nothing relevant.

A) A refusal and hand-off path <-- correct

B) A raised temperature setting

C) A larger context window

D) A fine-tuning of the model

Answer: A. Designing in a way for the system to say it does not know is one of the more effective defences, though where the threshold sits is a judgement.

Q34. Summarize the societal implications of Artificial Intelligence named on the syllabus.

A) Effects on the price of computing hardware only

B) Effects on the speed of internet connections only

C) Effects on the size of software companies only

D) Effects on employment, misinformation and equal treatment <-- correct

Answer: D. The syllabus places the ethical and societal implications alongside the technical material precisely because deployment decisions carry both.

Q35. Describe what the constraint section of a prompt contributes.

A) It supplies the documents to be searched

B) It bounds the length, scope and style of the reply <-- correct

C) It sets the number of parameters used

D) It selects the hardware the model runs on

Answer: B. Stating what the answer must not do, and how long it may be, removes a large share of the faults in a first attempt at a prompt.

Q36. Explain why the output of a generative model should be labelled as machine generated.

A) The label improves the accuracy of the output

B) The label is required by the model's licence

C) A reader is entitled to know how the content arose <-- correct

D) The label reduces the cost of each query

Answer: C. Disclosure is the transparency principle applied to generated content, and it bears directly on misinformation and on informed consent.

Q37. Distinguish Retrieval-Augmented Generation from simply training the model on more data.

A) Retrieval permanently alters the model's weights

B) Both of the two permanently alter the weights

C) Retrieval fetches sources at answer time without retraining <-- correct

D) Neither of the two changes what the model can cite

Answer: C. Retrieval keeps the knowledge outside the model, so it can be updated, audited and cited. Training bakes it in, where none of those things is possible.

Q38. Predict the difficulty of auditing a model whose internal reasoning cannot be inspected.

A) The model will refuse to produce any output

B) A prohibited factor may be in use without being visible <-- correct

C) The model cannot be deployed on any hardware

D) The model becomes accurate but very slow

Answer: B. Without an explanation there is no way to show that a decision did not turn on a factor that is not permitted, which is why explainability is a governance requirement.

Q39. Explain what the transparency principle requires of an organisation deploying an Artificial Intelligence system.

A) That the source code is published without charge

B) That the training data is sold to other firms

C) That the model is retrained every single month

D) That people are told the system is in use and how it decides <-- correct

Answer: D. Transparency covers disclosure that automation is being used, the basis on which it decides, and a route by which a person may question an outcome.

Q40. Interpret the finding that a hiring model scores candidates from one university consistently lower than equally qualified candidates elsewhere.

A) The model has learned a bias present in past decisions <-- correct

B) The model has been trained for too many epochs

C) The model's context window is set too small

D) The model's temperature has been set too high

Answer: A. The training records carry the decisions people made before, including the unfair ones, and the model reproduces the pattern that produced them.

----- End of Sample Question Set -----